Mini-project Task

2025-26

Mathematics Club

Centre for Innovation
IIT Madras

Instructions

§1 General

- Provide descriptive answers for all questions. Explain your thought process well.
- For the coding problems, here is the link to starter notebook. It is in depth so that you can work on the task easily without having any Python background. Links to documentations are also provided and you can always just google search. Avoid using LLMs unless really necessary.
- The instructions to use notebook are provided in the notebook. Add the link your version of it in the submission PDF.
- You are free to use the internet and any other resources for solving. These tasks are simply for you to get a feel of what the project will be like and learn as much as possible, it will not be graded in any manner. Resources and links for reading have been provided as well to ease your learning process. All the best and have fun!

§2 Submission

- Your solutions can be written manually, in digital notes, documents, or in LaTeX.
- The PDF should be named `<Firstname>_<ROLLNUMBER(IN CAPS)>.pdf` (For example: `Anand_MC25H007.pdf`).
- The final PDF is to be submitted in this Google form.
- The deadline for submission is **13 January, 2026, 23:59**.

§3 Contact Us

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Stochastic Optimization and Learning theory in STatIstiCal modEls (SOLSTICE)

Bhavi Goyal, Smitali Bhandari

Before we break down the project name, let me bring back some memories. Back in middle school, there used to be a Geography text book with theory and a note book with question and answers. "Learning" meant mugging up the set of questions and answers hoping that a subset of them would appear in the examination. Suppose that there is some content not in the notebook, but in the textbook. You only read the notebook, but the way you read it you are able to relate and answer questions from this content. This means, you are "generalizing"!

What is "Learning" in reality? How does your brain with billions of neurons or an AI model with trillions of neurons learn? We know what either of these learn from - data. Comparing with the previous analogy- does "generalization" make sense to you now? ChatGPT was possibly never trained on the exact question you ask it, but it can answer. You possibly never memorized that $2 + 2 = 5$, but see, you know that it is 5.

In the course of this project, we will first understand how to mathematically model these models as (Statistical) models. How do they do (Bayesian) Learning and how do we study these? Some tools we will need are from Stochastic Optimization.

Too much jargon, but as you will see it turns out to be pretty intuitive.

§1 Baye - sics

From the Mini-project Quest, you kind of already know this. There is a true distribution $q(x)$ from which is sampled $D = \{x_1, x_2, x_3, \cdots\}$. That is x_i follow the probability distribution q. As you know, any probability distibution can be determined from its parameters. For example, the parameters for a Normal Distribution are its mean μ and variance σ^2. So, when I say the true distribution is unknown, but the data D is known it means that given some n samples $x_1, x_2, \cdots x_n$ generated from some distribution. Can you determine the parameters governing the distribution?

In Quest, you were given a prior about the distribution of parameter p (β(1,1)). "Flip the coin until two heads" also gives you the probability density function of how the observation at 7th toss occurs. So from the Bayes' rule

$$P(w|D) \propto p(D|w) \cdot p(w_0)$$

you knew both the terms on LHS and hence w could be found using Bayesian estimate.
What if neither of them are given. Just the generated data. This is the real world scenario. How do we solve this?

Change the strategy-

$$\text{Choose method} \rightarrow \text{Result} \rightarrow \text{Evaluate chosen method}$$

It might seem that such an evaluation is impossible because we do not have any information about the true distribution. However, there are mathematical laws which hold for an arbitrary set of a true distribution, a statistical model, and a prior. By using formulas derived from the mathematical laws, we can evaluate the appropriateness of the set of a statistical model and a prior even if a true distribution is unknown.

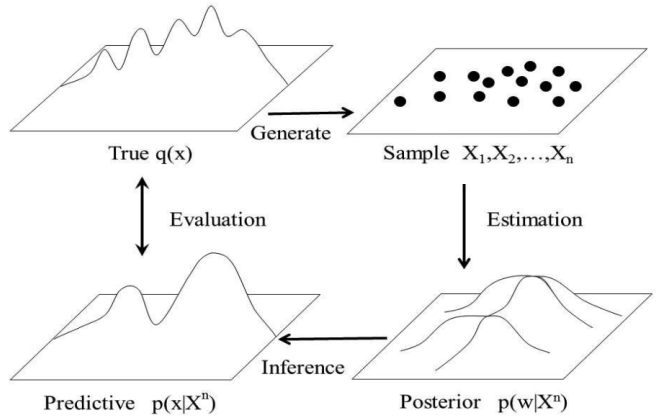

Figure 1.1: Framework of Bayesian inference. The procedure of Bayesian estimation is shown. A sample X^n is taken from unknown true distribution $q(x)$. A statistician sets a statistical model and a prior, then the posterior density $p(w|X^n)$ is obtained. The true distribution $q(x)$ is estimated by a predictive density $p(x|X^n)$, whose accuracy is evaluated by using mathematical laws.

Source: Mathematical Theory of Bayesian Statistics by Sumio Watanabe

A statistician who does not know $q(x)$, sets a statistical model $p(x|w)$ and a prior $\varphi(w)$, where $p(x|w)$ is a conditional probability density of x for a given w and $\varphi(w)$ is a probability density of w. (So we treat W on $\mathbb{R}^{\#\text{ of parameters}}$ also as a random variable and its realizations or samples are possible values of the parameter w (of dimension # of parameter(2 for a normal distribution μ, σ))). These $p(x|w)$ and $\varphi(w)$ maybe very incorrect for the true distribution as the statistician does not know it. The posterior probability density function of w is defined by

$$p(w \mid x_1, x_2, \ldots, x_n) = \frac{\prod_{i=1}^{n} p(x_i \mid w)\, \varphi(w)}{\int \prod_{i=1}^{n} p(x_i \mid w)\, \varphi(w)\, dw}$$

and the estimated probability density function of x is

$$\hat{p}(x) = \int p(x \mid w)\, p(w \mid x_1, x_2, \ldots, x_n)\, dw$$

also known as the predictive for the true unkown $q(x)$. The goal we wanted to achieve.
We will take a simple example and simulate this process for a simple setting.

Problem 1.1. Suppose the data D_n sampled from a true distribution $q(x)$ is

$$D_{10} = [-1.24, 0.35, 0.89, -0.47, 1.52, -0.98, 0.12, -0.31, 0.67, -0.55]$$

where n is the number of samples (trials of random variable X). What is the simplest true distribution $q(x)$ you can think of?

Problem 1.2. Smitali is too dumb and she could not realize what the true distribution can be and decided to follow the Bayesian Inference framework. She says, suppose that $w = (a, \sigma)$, my statistical model is conditioned on a normal distribution

$$p(x \mid a, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-a)^2}{2\sigma^2}\right)$$

and my prior is

$$\varphi(a, \sigma) = \begin{cases} \frac{1}{4}, & |a| < 1,\ 0 < \sigma < 2, \\ 0, & \text{otherwise.} \end{cases}$$

(Equal probability for any point in that 2D rectangular grid).

Computationally (you can write a python code to simulate this using the notebook) find $p(x_i|a, \sigma)$ for each $x_i \in D_{10}$. Now finding $p(D_{10}|a, \sigma)$ for each of a, σ means so much multiplication. Can you convert the multiplication into addition? Do that and find something which is proportional to $p(D_{10}|a, \sigma)$. (Hint: Vary a, σ in the said prior 2D rectangle for some 200 points, so a 200×200 grid.)

Problem 1.3. Multiply the obtained likelihood measure with the prior (If you followed the hint, you probably already did this as everything outside the rectangle has a probability zero.) and normalize to obtain $p(a, \sigma|D_{10})$ for every a, σ in the grid.

Problem 1.4. (Continue the same code) Use your knowledge of expectation and marginal probabilities, joint probabilites to estimate $\hat{a}, \hat{\sigma}$ and hence the estimated distribution of X $\mathcal{N}(\hat{a}, \hat{\sigma})$. Compare this with what you naively guessed. How dumb was Smitali to do this tedious inference method?
(Hint:

```
#example code block
print("Smitali is dumbest")
✓ 0.0s

Smitali is dumbest
```

)

§2 The Art of Guessing: Estimators

We have seen how to find the posterior via a grid. In practice, we use **Estimators**—rules that take data D and provide a specific "best" w.

§2.1 Point Estimates vs. Predictive Distributions

The **Maximum Likelihood Estimator (MLE)** is the w that maximizes the likelihood $p(D|w)$. However, a Bayesian prefers the **Predictive Distribution**, which accounts for the entire landscape:

$$\hat{p}(x) = \int p(x \mid w) p(w \mid D) dw$$

Problem 2.1. Using your code for the Normal Distribution from Section 1:

1. Identify the $(\hat{a}, \hat{\sigma})$ that gives the maximum probability (the MLE).
2. Plot $p(x|\hat{a}, \hat{\sigma})$ and overlay it with the Bayesian predictive $\hat{p}(x)$.
3. **Reflection:** With a small dataset ($n = 3$), which estimate is more robust to a single outlier?

§3 The n in D_n matters!

Note: Please use the notebook provided for these questions.

1. Assuming your guess of a, σ from 1.1 to be the true $q(x)$, generate 10 datasets D_n with sample size 10. Using $p(a, \sigma|D_{10})$ for each ten of the $D'_{10}s$ plot a contour map for a, σ. You can use the same prior and model as Smitali if you wish to. Brownie points for trying various other intutive prior + model combinations. Also mark the true a, σ that you assumed in each plot. What can you conclude from the plots?
2. Do the same for $n = 20, 30$. What can you conclude?

§4 Food for thought

This section does not expect perfect answers. Explore and if you have something to share/ ask, feel free to include in your submission. There might be some unknown terms below, here are some starting resources:

- 3b1b Neural Network
- Gradient Descent - How Neural Networks learn
- Welch labs (really nice channel for math in ai)
- Edmund's resources on SLT. The basics are very well explained.
- Pretty intutive Introduction to SLT by Sumio Watanabe
- Algebraic Geometry and Statistical Learning Theory This is a bit advanced, skip for now, but if interested, go ahead :)

Problem 4.1. Section 1 was simply 10 samples sampled from a true normal distribution. In real world, the true distribution is definitely not normal and definitely not without noise (We were too good to not add any noise in your D_{10}). Real world like the common crawl of internet is contaminated with noise. This is the dataset for many models. You saw that sampling itself adds noise. And then the real world noise. How do you surpass these noise and make your model "generalize" to the population's (true) parameters and not just sample estimated parameters?

Problem 4.2. We just had one guess, we found the likelihood and then the predictive. How do I do the evaluation step? How do I update my belief (prior)?

Problem 4.3. Models like Linear Regression, Neural networks, start with some initialization of their parameters and update these values as they see more data (get more trained). What optimization methods do they use for this? How similar or different it is to being Bayesian Update?

Problem 4.4. If they are different can we still study them in Bayesian way?

Problem 4.5. Suppose I give you data such that multiple statistical models are correct for it. That is the relation is not one-to-one. What are these known as? Is this relation one to one